Filtering the Back End of a Futures Curve from the Liquid Front

1. Setup

Contracts $j = 1, \dots, J$ have fixed expiries T_j; $\tau_j(t) = T_j - t$. Let $y_j(t) = \ln F_j(t)$ be the latent *true* log futures price. Observations arrive at event times $t_1 < t_2 < \cdots$ (the union of all trade and quote-revision timestamps), with $\Delta_k = t_k - t_{k-1}$. At t_k only the subset $\mathcal{O}_k \subseteq \{1, \dots, J\}$ of contracts that printed or revised a quote in $(t_{k-1}, t_k]$ is observed; $n_k = |\mathcal{O}_k|$, and $W_k \in \{0,1\}^{n_k \times J}$ selects those rows.

2. State Dynamics

Under $\mathbb{P}$, with $d\langle W^\chi, W^\xi \rangle_t = \rho\, dt$ (Schwartz–Smith, 2000), plus a back-end factor u_t:

$$d\chi_t = -\kappa\chi_t\, dt + \sigma_\chi dW_t^\chi, \qquad d\xi_t = \mu_\xi\, dt + \sigma_\xi dW_t^\xi, \qquad du_t = -\kappa_u u_t\, dt + \sigma_u dW_t^u, \quad W^u \perp (W^\chi, W^\xi).$$

For $\alpha_k = (\chi_{t_k}, \xi_{t_k}, u_{t_k})^\top$ the exact discretization is $\alpha_k = c_k + \Phi_k \alpha_{k-1} + \eta_k$, $\eta_k \sim N(0, Q_k)$, with

$$c_k = \begin{pmatrix} 0 \\ \mu_\xi \Delta_k \\ 0 \end{pmatrix}, \quad \Phi_k = \operatorname{diag}\big(e^{-\kappa\Delta_k},\, 1,\, e^{-\kappa_u \Delta_k}\big),$$

$$Q_k = \begin{pmatrix} \frac{\sigma_\chi^2}{2\kappa}(1 - e^{-2\kappa\Delta_k}) & \frac{\rho\sigma_\chi\sigma_\xi}{\kappa}(1 - e^{-\kappa\Delta_k}) & 0 \\ \cdot & \sigma_\xi^2 \Delta_k & 0 \\ 0 & 0 & \frac{\sigma_u^2}{2\kappa_u}(1 - e^{-2\kappa_u\Delta_k}) \end{pmatrix}.$$

Initialization: $\chi_0 \sim N(0, \sigma_\chi^2/2\kappa)$, $u_0 \sim N(0, \sigma_u^2/2\kappa_u)$, ξ_0 diffuse.

3. Measurement Equation

With market prices of risk $\lambda_\chi, \lambda_\xi$ and $\mu_\xi^* = \mu_\xi - \lambda_\xi$, Schwartz–Smith pricing gives $y_j(t) = e^{-\kappa\tau}\chi_t + \xi_t + A(\tau)$, $\tau = \tau_j(t)$, where

$$A(\tau) = \mu_\xi^* \tau - \frac{\lambda_\chi}{\kappa}(1 - e^{-\kappa\tau}) + \tfrac{1}{2}\Big[\frac{\sigma_\chi^2}{2\kappa}(1 - e^{-2\kappa\tau}) + \sigma_\xi^2 \tau + \frac{2\rho\sigma_\chi\sigma_\xi}{\kappa}(1 - e^{-\kappa\tau})\Big].$$

We extend this reduced-form with the back-end factor and a monthly seasonal $s_{m(j)}$ ($\sum_m s_m = 0$):

$$y_j(t_k) = d_{kj} + z_{kj}\, \alpha_k, \qquad z_{kj} = \big(e^{-\kappa\tau_{kj}},\, 1,\, g(\tau_{kj})\big), \qquad d_{kj} = A(\tau_{kj}) + s_{m(j)}, \qquad g(\tau) = 1 - e^{-\tau/\tau_u}.$$

Since $g(\tau) \approx 0$ for front contracts, u_t is observable only through the back end: it carries back-end-specific news (long-dated supply, hedging flow, term premium shifts) that the two-factor model would wrongly attribute to (χ, ξ). Stacking, $y_k = d_k + Z_k \alpha_k$, and the observation is

$$y_k^o = W_k\big(d_k + Z_k\alpha_k + \varepsilon_k\big), \qquad \varepsilon_k \sim N(0, H_k), \quad H_k = \operatorname{diag}(h_{k1}, \dots, h_{kJ}).$$

Liquidity-dependent noise. With log bid–ask spread $S_{kj} = \ln(\text{ask}_{kj}/\text{bid}_{kj})$ and $\mathbb{1}^{\text{tr}}_{kj}$ indicating a trade (vs. a quote),

$$h_{kj} = \omega^2 + (1 - \mathbb{1}^{\text{tr}}_{kj})\, S_{kj}^2/12 + \psi \mathbb{1}^{\text{roll}}_{kj},$$

where $S^2/12$ is the variance of a true price uniform on the quoted interval and $\mathbb{1}^{\text{roll}}$ flags contracts within the roll/expiry window.

Staleness. On the event-time grid, an unrevised quote is simply not in $\mathcal{O}_k$, so stale prices never enter the filter. If instead data are sampled on a fixed grid (e.g. daily closes) with last-update time $\ell_{kj} \le t_k$, either (i) drop j from $\mathcal{O}_k$ when $\ell_{kj} \le t_{k-1}$, or (ii) insert ℓ_{kj} into the grid so the price is matched to the state at the time it was set. Approximating staleness by inflating h_{kj} by $\gamma(t_k - \ell_{kj})$ is a cruder alternative.

Spreads and strips. A calendar spread $y_i - y_j$ enters as a row $(z_{ki} - z_{kj},\, d_{ki} - d_{kj})$ with its own noise variance. A strip price $\bar{F} = \frac{1}{|S|}\sum_{j\in S} F_j$ enters, to first order in the dispersion of $\{y_j\}_{j\in S}$, as the row $\frac{1}{|S|}\sum_{j\in S}(z_{kj}, d_{kj})$.

4. Filter, Likelihood, and Marks

Let $\tilde{Z}_k = W_k Z_k$, $\tilde{d}_k = W_k d_k$, $\tilde{H}_k = W_k H_k W_k^\top$. For $k \geq 1$:

Predict:	$a_{k\mid k-1} = c_k + \Phi_k a_{k-1\mid k-1}$,	$P_{k\mid k-1} = \Phi_k P_{k-1\mid k-1} \Phi_k^\top + Q_k$,
Innovate:	$v_k = y_k^o - \tilde{d}_k - \tilde{Z}_k a_{k\mid k-1}$,	$F_k = \tilde{Z}_k P_{k\mid k-1} \tilde{Z}_k^\top + \tilde{H}_k$,
Update:	$a_{k\mid k} = a_{k\mid k-1} + K_k v_k$,	$P_{k\mid k} = (I - K_k \tilde{Z}_k) P_{k\mid k-1}$, $\quad K_k = P_{k\mid k-1} \tilde{Z}_k^\top F_k^{-1}$.

If $n_k = 0$, set $a_{k\mid k} = a_{k\mid k-1}$, $P_{k\mid k} = P_{k\mid k-1}$. With the diffuse component handled by exact diffuse initialization (Durbin–Koopman, Ch. 5), the log-likelihood is $\ell(\theta) = -\frac{1}{2} \sum_{k:\, n_k > 0} \left[n_k \ln 2\pi + \ln |F_k| + v_k^\top F_k^{-1} v_k \right]$.

Proposition. *For any contract b, the filtered mark $\hat{y}_{kb} = d_{kb} + z_{kb} a_{k\mid k}$ satisfies*

$$\hat{y}_{kb} - \hat{y}_{kb}^- = \mathrm{Cov}_{k-1}\big(y_b(t_k),\, y_k^o\big)\, F_k^{-1} v_k = z_{kb} P_{k\mid k-1} \tilde{Z}_k^\top F_k^{-1} v_k,$$

where $\hat{y}_{kb}^- = d_{kb} + z_{kb} a_{k\mid k-1}$, i.e. the back-end revision is the Gaussian regression of the back on the front innovations. If only one front contract f is observed, $\hat{y}_{kb} - \hat{y}_{kb}^- = \dfrac{z_{kb} P_{k\mid k-1} z_{kf}^\top}{z_{kf} P_{k\mid k-1} z_{kf}^\top + h_{kf}}\, v_{kf}$.

Proof. Joint Gaussianity of $(y_b(t_k), y_k^o)$ given $\mathcal{F}_{k-1}$, with $\mathrm{Cov}_{k-1}(z_{kb}\alpha_k, \tilde{Z}_k \alpha_k + W_k \varepsilon_k) = z_{kb} P_{k\mid k-1} \tilde{Z}_k^\top$. □

Mark at arbitrary time $t^* \in [t_k, t_{k+1})$. With $\Delta = t^* - t_k$, $a_{t^*} = c(\Delta) + \Phi(\Delta) a_{k\mid k}$ and $P_{t^*} = \Phi(\Delta) P_{k\mid k} \Phi(\Delta)^\top + Q(\Delta)$. The mark for contract b is $\hat{y}_b(t^*) = d_b(t^*) + z_b(t^*) a_{t^*}$, with variance $z_b(t^*) P_{t^*} z_b(t^*)^\top$ for the true price and an additional h_b for the next print.

Why the third factor matters. If at least two front contracts with distinct maturities are observed, the rows $(e^{-\kappa\tau}, 1)$ are linearly independent, so (χ, ξ) is observable and its covariance stays bounded. Without u_t, the back-end mark variance therefore does *not* grow with time since the last back trade, which is too confident. With u_t, $[P_{t^*}]_{uu}$ rises toward $\sigma_u^2 / 2\kappa_u$ between back-end prints. Its contribution to the mark variance is $g(\tau_b)^2 [P_{t^*}]_{uu}$, so the band widens with staleness at the long end only.

5. Estimation and Identification

The parameter vector $\theta = (\kappa, \sigma_\chi, \sigma_\xi, \rho, \mu_\xi, \lambda_\chi, \mu_\xi^*, \kappa_u, \sigma_u, \tau_u, s_{1:11}, \omega, \psi)$ is estimated by $\hat{\theta} = \arg\max \ell(\theta)$ under the constraints $\kappa, \kappa_u, \sigma_\cdot, \tau_u, \omega > 0$ and $|\rho| < 1$. Points to note: (i) μ_ξ is identified only from the time series and is weak; the cross-sectional drift μ_ξ^* is well identified from the slope of the curve. (ii) κ enters the loadings $e^{-\kappa\tau}$, so the back end extrapolates from the front through it. Its estimate is dominated by the liquid maturities. (iii) $(\kappa_u, \sigma_u, \tau_u)$ are identified only from genuine back-end prints. Fit them on a clean subsample, or put priors on them. (iv) Exchange settlements for back months are often derived from front prices plus spreads. Excluding them avoids feeding the model's own extrapolation back as data.

6. Validation

For each genuine back-end print (b, t_k), form the one-step standardized error $e_{kb} = (y_{kb}^o - \hat{y}_{kb}^-) / (z_{kb} P_{k\mid k-1} z_{kb}^\top + h_{kb})^{1/2}$. Under the model, the e_{kb} are i.i.d. $N(0,1)$. Test this with a Ljung–Box test and a normal Q–Q plot, and check interval coverage as a function of time since the last print. Compare the RMSE of $y_{kb}^o - \hat{y}_{kb}^-$ with two benchmarks: (a) the last traded price and (b) the last price plus $\hat{\beta}\times$ the front move since then, with $\hat{\beta}$ estimated by OLS on synchronous changes.

References. Schwartz, E. and J. Smith (2000), "Short-term variations and long-term dynamics in commodity prices," *Management Science* 46(7). Durbin, J. and S. J. Koopman (2012), *Time Series Analysis by State Space Methods*, 2nd ed., Oxford.